Ex.No: 8 Implement Packet shiffing using Raw Sockets in Python

Step 1: Install python.
Download and install python from the official python website
Select add python to path during installation.
Verify installation:
python --version

Step 2: Install vs code
Download and install vs code
Install the python extension.

Step 3: Verify python
open command prompt and type
python
print("Hello")

Step 4: Create project Folder
Create a Folder named PacketShiffer
Inside the Folder, Create:
shiffer.py

Step 5: Write the python program

```
import socket
import struct
shiffer = socket.socket(socket.AF_INET, socket.SOCK_RAW,
                        socket.IPPROTO_IP)
host = socket.gethostbyname(socket.gethostname())
shiffer.bind((host,0))
shiffer.setsockopt(socket.IPPROTO_IP,
                   socket.IP_HDRINCL,1)
shiffer.ioctl(socket.SIO_RCVALL,socket.RCVALL_ON)
print("Packet shiffer started...")
try:
    while True:
```

```
raw_data, addr = sniffer.recvfrom(65535)
ip_header = raw_data[0:20]
iph = struct.unpack("!BBHHHBBH4s4s", ip_header)
version = iph[0] >> 4
ihl = iph[0] & 0xF
ttl = iph[5]
protocol = iph[6]
source = socket.inet_ntoa(iph[8])
destination = socket.inet_ntoa(iph[9])
print("-----")
print("Version:", version)
print("Header Length:", ihl*4)
print("TTL:", ttl)
print("Protocol:", protocol)
print("Source IP:", source)
print("Destination IP:", destination)
print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("Stopped")
```

Step 6: Save the Program

Save the file as:

Sniffer.py

Step 7: Run command prompt as Administrator.

Search command prompt.

Right-click → Run as Administrator.

Step 8: Navigate to the Folder.

cd Desktop

cd Packet Sniffer

Step 9: Run the Program

python sniffer.py.

Expected output:

The program displays captured network packet details such as:

Source IP address.

Destination IP address.

Protocol

TTL

Packet length.

Result:

Thus, the packet sniffing program using RAW sockets in python was successfully implemented and executed.